Problem Statement

My client is an economist, working for the World Bank. She struggles to access data relating to world development indicators from the official World Bank website and has requested an application to make it easier for her to access the data.

Scenario

The client is my sister, an economist working for the World Bank. Her job involves traveling to many countries and helping formulate policies for said countries. In the countries she travels to, she often needs to know about the country's development and economic trends, policies, and indicators, all of which are available in an open database, on the World Bank Website. However She often struggles with this due to it being extremely hard to use, and an unfriendly user interface. As a result, we had an interview which can be found in Appendix A. to develop an application that would allow her to easily access the database, and provide a list of options for data she could select, from a timeframe of this current year, five years, and ten years.

Over the course of the interview, she suggested some features that'll aid in her job as an economist, such as the ability to opt in, for notifications, which give out economic facts every 24 hours about the country she is in at the moment. Moreover, she gave specific data sets she would prefer such as Maternal Mortality, School Enrollment Level, Health Access, Water and Sanitation, rate of urbanization, GDP, Exports, and Imports, and Revenue from taxes and their expenditures are the prioritized indicators.

For a start, we settled with a desktop application. A desktop application would be running from the launch of the operating system, and would provide daily notifications about economic data.

Possible Solutions

1. A possible solution would be to create an app which sends an email each time the client gets to a new country, which would contain the desired info.
2. Another was a mobile application, that consistently sends information about the country
3. A third was a desktop application, that could have the clients desired data, with which they could access, create tables and visualize.

Rationale For Proposed Product

I decided to go for the third option. This application will be developed with Java, as thanks to the Java Virtual Machine, it allows for compatibility with multiple platforms, as long as JVM is available for it. There are also a lot of frameworks and tools available, especially when working with DataBases, as there is a lot of support for connecting to databases.

I specifically used Netbeans to program this application in Java,, as this would allow for easy UI development. Netbeans also allows for easy integration with Java and its APIs, one of which will allow me to obtain location data. Not to mention

To implement the required database of World Development indicators, I will use MongoDB, as it provides an easy, and user-friendly way to manage large sets of data. I chose this over MySQL, as MongoDB provides a tool for visualizing data which will make implementing graphs much easier.

Success Criteria.

1. Application automatically detects users location
2. Application allows user to change their country to override automatic location dector
3. Application's main screen provides general information about the country the user is in.

4. Application detects, when user is in a new country and adjusts data accordingly
5. Application can be navigated with tabs to various sections (Dashboard, Database, Settings)
6. Application allows user to retrieve specific world development indicator data of a country from a given year
7. Application allows user to insert data from a given time period on to a table
8. Table can be manipulated(Redo, Undo, and Clear functionality)
9. Program allows for visualization of given data using a graph.
10. Program allows for graphs to be saved as an image file
11. Notifications are sent, based on facts from the database about the users country
12. Notifications can be localized to users specific country, or provide information about different countries